

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 7 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|---------------------------------------|--|
| 1. Singular value _____ | (A) A column times a row, $\mathbf{uv}^T$ — a whole matrix built from two vectors. |
| 2. Singular vectors _____ | (B) The factorization $A = U\Sigma V^T$ (rotate–stretch–rotate) that <i>every</i> matrix has. |
| 3. Singular Value Decomposition _____ | (C) A number $\sigma \geq 0$: how far A stretches along a singular direction (a half-width of the ellipse). |
| 4. Outer product _____ | (D) A square matrix with perpendicular unit columns, so $Q^T Q = I$ and $Q^{-1} = Q^T$. |
| 5. Rank-one matrix _____ | (E) The perpendicular unit inputs $\mathbf{v}_i$ and outputs $\mathbf{u}_i$ with $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$. |
| 6. Orthogonal matrix _____ | (F) A matrix whose every column is a multiple of one vector (like $\mathbf{uv}^T$). |
| 7. Principal component _____ | (G) The total $\sigma_1^2 + \sigma_2^2 + \dots$, equal to the sum of the squares of all entries of A . |
| 8. Energy of a matrix _____ | (H) A top singular direction of centered data — the direction of greatest spread. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- The singular values σ of a matrix are (A) always ≥ 0 (B) sometimes negative (C) the eigenvalues of A (D) always 1
- To find the singular values of A , use the eigenvalues of (A) A (B) $A + A^T$ (C) $A^T A$ (D) A^{-1}
- A singular value σ_i and the eigenvalue λ_i of $A^T A$ are related by (A) $\sigma = \lambda$ (B) $\sigma = \sqrt{\lambda}$ (C) $\sigma = \lambda^2$ (D) $\sigma = 1/\lambda$
- In $A = U\Sigma V^T$, all the stretching is carried by (A) U (B) V (C) Σ (D) V^T
- An orthogonal matrix Q satisfies (A) $Q^2 = I$ (B) $Q^T Q = I$, so $Q^{-1} = Q^T$ (C) $\det Q = 0$ (D) $Q = Q^T$ always
- To compress an image with the SVD, you keep the rank-one layers with the (A) smallest σ (B) largest σ (C) largest $\mathbf{u}$ (D) negative σ

Part C — Short Answer & Computation (35 pts)

Show your work.

1. **(a)** The diagonal matrix $D = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ acts on the unit circle. What are its singular values $\sigma_1 \geq \sigma_2$, and what are the half-widths of the ellipse it makes? **(b)** Find the singular values of the swap $S = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$.
2. Find the **singular values** of $A = \begin{bmatrix} 2 & 2 \\ 1 & -2 \end{bmatrix}$ by forming $A^T A$, finding its eigenvalues, and taking $\sigma = \sqrt{\lambda}$.
3. For that same $A = \begin{bmatrix} 2 & 2 \\ 1 & -2 \end{bmatrix}$, the input vector for $\sigma_1 = 3$ is $\mathbf{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Find the **output singular vector** $\mathbf{u}_1 = \frac{A\mathbf{v}_1}{\sigma_1}$, and check that it is a unit vector.
4. **(a)** Compute the outer product $\mathbf{u}\mathbf{v}^T$ where $\mathbf{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. **(b)** Explain why the result is a **rank-one** matrix.
5. The symmetric matrix $A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$ has singular values $\sigma_1 = 6$, $\sigma_2 = 2$ with $\mathbf{u}_1 = \mathbf{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. **(a)** Build the best rank-one approximation $A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$. **(b)** What fraction of the **energy** does A_1 keep?
6. Let $Q = \frac{1}{5} \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$. **(a)** Show Q is **orthogonal** by checking $Q^T Q = I$. **(b)** Write down Q^{-1} . **(c)** The vector $\mathbf{x} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ has length 5. Find $Q\mathbf{x}$ and its length, and explain.

7. Four data points are $(5, 7)$, $(7, 5)$, $(3, 1)$, $(1, 3)$. **(a) Center** the data (subtract the mean). **(b)** Form $A^T A$. **(c)** Find its eigenvalues and the first principal component PC_1 . **(d)** What fraction of the spread does PC_1 capture?

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. *Every* matrix has an SVD, even though not every matrix has real eigenvectors. **(a)** Explain why $A^T A$ is symmetric for *any* matrix A , and why that guarantees its eigenvalues are real with $\lambda \geq 0$ and its eigenvectors are perpendicular. **(b)** Explain why this makes $\sigma = \sqrt{\lambda}$ a real number ≥ 0 , so every matrix has singular values.
2. The SVD $A = U\Sigma V^T$ is “rotate–stretch–rotate.” **(a)** Explain what each of the three factors does geometrically, and why *all* the stretching lives in Σ . **(b)** An orthogonal matrix Q has $Q^T Q = I$. Explain why this makes $Q^{-1} = Q^T$ (a free inverse) and why Q preserves length.